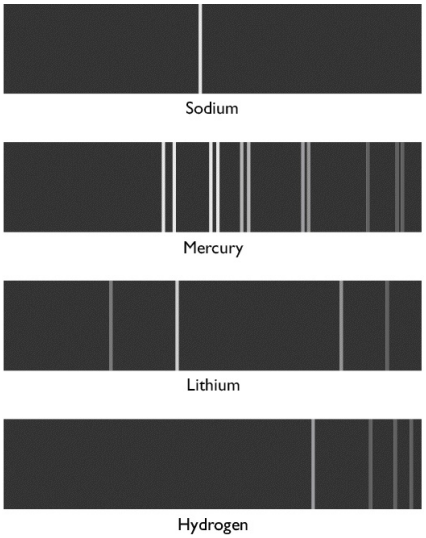


Figure 4.2 The spectra of some elements: sodium, mercury, lithium and hydrogen.

Colour is the speed at which Faraday's lines vibrate, and this is determined by the vibrations of the electric charges which emit light. These charges are the electrons that move inside the atoms. Therefore, studying spectra, we can understand how electrons move around nuclei. The other way around, we could predict the spectrum of each atom by computing the frequencies of the electrons circling their nucleus. Easy to say, but in practice nobody was able to do so. In fact, the whole thing seemed impossible, because in Newton's mechanics an electron can revolve around its nucleus at *any* speed, and hence emit light at *any* frequency. But then why does the light emitted by an atom not contain *all* colours, rather than just a few particular ones? Why are atomic spectra not a continuum of colours, instead of just a few separate lines? Why, in technical parlance, are they 'discrete' instead of continuous? For decades, physicists seemed incapable of finding an answer.

Bohr finds a tentative solution, by way of a strange hypothesis. He realizes that everything could be explained if the energy of electrons in atoms could only assume certain 'quantized' values – certain specific values, just as was hypothesized by Planck and by Einstein for the energy of the quanta of light. Once again, the key is a *granularity*, but not now for the energy of light but rather for the energy of the electrons in the atom. It begins to become clear that granularity is something widespread in nature.

Bohr makes the hypothesis that electrons can exist only at certain 'special' distances from the nucleus, that is, only on certain particular orbits, the scale of which is determined by Planck's constant *h*. And that electrons can 'leap' between one orbit with the permitted energy to another. These are the famous 'quantum leaps'. The frequency at which the electron moves on these orbits determines the frequency of the emitted light and, since only certain orbits are allowed, it follows that only certain frequencies are emitted.